

ROSEAU, MN, USA

WMO: 727477

Lat: 48.856N

Lon: 95.697W

Elev: 1059

StdP: 14.14

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04970

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.3	-18.3	-31.1	1.0	-23.0	-25.8	1.4	-18.9	25.3	14.8	22.9	13.4	4.8	270	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	87.7	74.6	83.7	71.5	81.3	69.6	76.3	83.8	74.0	81.4	71.5	78.7	11.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	128.7	81.2	71.9	122.7	79.9	69.7	113.4	77.6	40.5	84.1	38.2	80.8	36.0	77.9	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.8	20.4	18.4	DB	-29.8	91.6	6.1	3.4	-34.2	94.0	-37.8	96.0	-41.2	97.9	-45.6	100.4	(4)
			WB	-30.9	80.1	6.5	3.4	-35.6	82.6	-39.4	84.6	-43.1	86.5	-47.8	89.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.7	6.2	10.2	24.3	40.2	53.0	63.3	68.0	65.9	58.0	44.3	28.5	12.9
	DBStd	24.20	15.39	13.72	13.73	10.46	8.69	6.73	5.63	5.97	8.75	8.86	11.52	13.75
	HDD50	5709	1358	1115	799	320	68	2	0	0	25	222	648	1152
	HDD65	9547	1823	1535	1262	743	382	109	34	62	241	643	1096	1617
	CDD50	1953	0	0	2	27	161	400	557	492	266	46	2	0
	CDD65	315	0	0	0	0	10	57	126	89	32	1	0	0
	CDH74	3254	0	0	0	12	162	568	1215	914	362	21	0	0
	CDH80	837	0	0	0	1	31	135	328	242	97	3	0	0
Wind	WSAvg	8.3	8.1	8.1	8.6	9.3	9.5	8.2	7.2	7.1	8.3	8.9	8.9	8.1
Precipitation	PrecAvg	20.9	0.7	0.5	0.6	1.4	2.5	4.1	3.3	3.2	2.5	1.5	0.7	0.7
	PrecMax	30.0	3.1	1.3	1.7	3.1	8.5	9.3	7.2	11.0	6.9	4.4	2.7	1.8
	PrecMin	14.4	0.0	0.0	0.0	0.0	0.2	0.6	1.1	0.7	0.3	0.1	0.0	0.0
	PrecStd	3.4	0.7	0.3	0.5	0.9	1.7	1.9	1.5	2.2	1.7	1.1	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	42.7	62.8	74.9	83.7	88.6	90.7	90.4	88.1	76.9	60.9	40.6
		MCWB	37.7	39.1	53.7	57.7	66.4	75.1	77.2	76.3	73.4	63.4	52.0	36.5
	2%	DB	36.0	37.3	52.1	68.5	79.0	83.6	86.4	85.7	82.2	69.9	53.6	36.7
		MCWB	34.1	35.4	46.2	53.6	61.1	71.2	75.1	72.3	68.8	59.3	48.1	34.6
	5%	DB	32.2	34.5	45.5	63.2	73.3	80.8	82.5	81.9	78.8	63.5	47.9	34.4
		MCWB	30.8	33.0	41.4	50.8	58.8	68.3	72.2	70.4	66.0	55.1	43.9	32.6
	10%	DB	27.7	30.4	41.0	57.1	69.9	76.9	81.1	79.4	73.0	57.4	43.5	30.5
		MCWB	26.3	29.6	37.8	47.6	57.7	65.9	71.1	68.6	63.4	51.1	39.5	29.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	39.2	54.2	59.1	69.9	78.4	80.3	78.9	75.2	65.5	53.3	37.4
		MCDB	40.3	41.2	59.6	70.7	79.0	84.8	87.4	88.1	85.1	71.7	58.6	39.5
	2%	WB	33.3	36.3	46.2	54.9	65.2	73.8	76.9	75.1	71.3	60.4	47.8	34.9
		MCDB	35.4	37.9	50.6	65.5	74.6	81.3	83.9	82.8	79.1	68.3	52.5	36.7
	5%	WB	30.6	33.3	41.4	51.7	61.7	70.4	74.6	72.5	67.9	55.5	44.3	32.5
		MCDB	31.9	34.9	45.2	61.5	69.6	77.2	81.4	80.2	75.7	61.6	47.8	34.3
	10%	WB	26.4	29.8	38.1	48.6	58.8	67.5	72.2	70.1	65.0	52.1	40.0	29.8
		MCDB	27.6	30.7	41.1	57.3	66.9	74.9	79.0	77.4	71.8	57.9	44.0	30.6
Mean Daily Temperature Range	5% DB	MDBR	17.4	18.6	18.1	19.9	21.5	19.7	21.0	22.4	22.1	17.4	14.4	15.1
		MCDBR	16.7	16.2	19.5	29.2	28.4	24.7	24.0	25.9	27.9	25.9	20.7	15.5
		MCWBR	15.3	14.5	14.9	17.4	15.1	14.3	15.2	15.7	16.5	16.4	15.2	14.1
	5% WB	MCDBR	16.3	15.5	19.1	26.3	23.5	21.1	22.0	23.2	24.3	23.8	19.2	15.2
		MCWBR	15.1	14.0	14.7	16.5	14.8	14.0	15.4	15.6	16.0	16.2	14.8	13.9
Clear-Sky Solar Irradiance	taub	0.265	0.300	0.294	0.344	0.361	0.377	0.381	0.375	0.347	0.303	0.276	0.259	
	taud	2.284	2.180	2.404	2.374	2.393	2.374	2.380	2.401	2.471	2.560	2.495	2.365	
	Ebn at Noon	254	267	294	287	285	280	277	274	272	270	253	243	
	Edh at Noon	23	32	30	35	36	37	36	34	29	22	19	18	
All-Sky Solar Radiation	RadAvg	445	828	1208	1525	1712	1807	1933	1607	1186	701	395	317	
	RadStd	49	65	118	136	111	142	90	115	93	101	48	35	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page